

Revisiting the Lévy Foraging Hypothesis in 2D with Reproducible Simulation

Thesis Pre-defence

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Background: Random Search

Sparse-target blind search (limited cues).

Lévy flights - heavy-tailed step-length random walk (power-law).

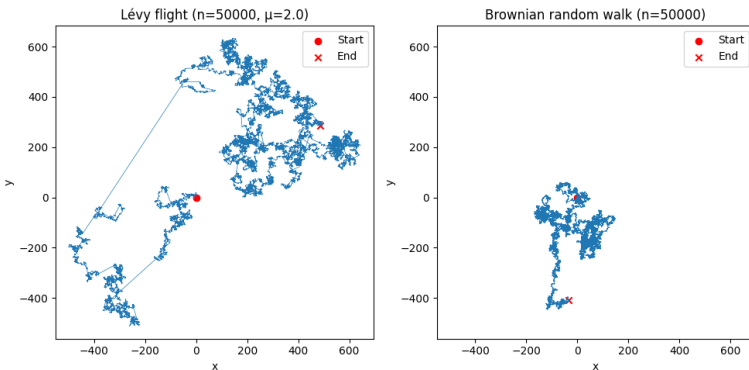


Figure: Lévy (Cauchy) and Brownian trajectories of different lengths.

Problem: the Lévy foraging hypothesis

Claim (Viswanathan et al., 1999): inverse-square Lévy flights ($\mu \approx 2$) maximise encounter rates in sparse environments.

Is $\mu^* = 2$ truly universal across densities and post-detection strategies, or is it an artifact of specific modeling assumptions?

Why it matters

- Biology: interpreting animal movement data and foraging optimality.
- Robotics: efficient exploration/search without maps.
- Theory: resolving contradictory “Lévy optimality” results.

Why timely

- Mature debate + high-quality critiques exist, but a **unified benchmark is missing**.

Lévy flights and the search model

Lévy flights: random walks with power-law distributed step lengths.

- $\mu \rightarrow 1$: ballistic (long straight flights)
- $\mu = 2$: inverse-square ("Lévy optimal")
- $\mu \rightarrow 3$: Brownian-like (diffusive)

Viswanathan model:

- 2D infinite plane, homogeneous Poisson targets at density ρ , detection radius r_v .
- Mean free path $\lambda = 1/(2\rho r_v)$; sparsity λ/r_v .
- Flight length: $p(l) \propto l^{-\mu}$, $l \in [l_{\min}, l_{\max}]$.
- Destructive vs. non-destructive post-detection strategies.

Goals and objectives

The aim of this work is to construct a reproducible 2D simulation benchmark and to determine quantitatively under which conditions the inverse-square Lévy exponent $\mu^* = 2$ is optimal, when the optimum shifts to ballistic or Brownian-like regimes, and how sensitively μ^* depends on the post-detection strategy, target sparsity, and truncation parameters.

Objectives:

- **Build a reproducible 2D simulator** with truncated power-law flights ($\mu \in [1, 3]$), destructive and non-destructive search.
- **Sweep key factors:** μ , target density ρ (via sparsity $\lambda/r_v \in [10, 2000]$), and four post-detection strategies.
- **Measure efficiency metrics:** pooled unique-capture rate η and MFPT; assess statistical convergence.
- **Construct the phase diagram** $\mu^*(\lambda/r_v)$ across all strategies.

Evidence Supporting Lévy Optimality

Multiple experimental and theoretical papers claim Lévy optimality: efficiency peaks at $\mu \approx 2$ for destructive search in sparse Poisson fields Viswanathan et al. (1999); Raposo et al. (2003); Bartumeus et al. (2002).

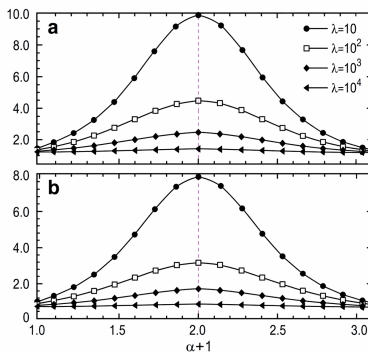


Figure: $\lambda\eta$ vs. μ (from Viswanathan et al., 1999). Peak near $\mu = 2$.

Contradictions and Context Dependence

Empirical re-analyses Edwards et al. (2007):

1. Likelihood-based model selection favors gamma or composite Brownian over pure power laws.
2. Apparent power-law behavior can arise from data pooling and segmentation.

Theoretical critiques:

1. Palyulin et al. (2014): drift/flows can make Brownian/ballistic better than Lévy for MFPT.
2. Levernier et al. (2020): in $d \geq 2$, encounter rate scales linearly with ρ for all μ ; optimality is a prefactor effect.
3. Buldyrev et al. (2021): reset distance l_c and destructive vs. non-destructive rules qualitatively change the optimum.

Optimal μ is highly sensitive to drift, boundaries, and microscopic reset rules.

Approach overview

1. 2D infinite plane with chunked PPP target generation.
2. Four post-detection strategies (destructive and non-destructive):
 - No interrupt, Teleport ($l_c = 5r_v$), Viswanathan destructive, Viswanathan non-destructive.
3. Sweep $\mu \in [1, 3]$ from ballistic ($\mu \rightarrow 1$) through Lévy ($\mu \approx 2$) to Brownian ($\mu \rightarrow 3$) at each density.

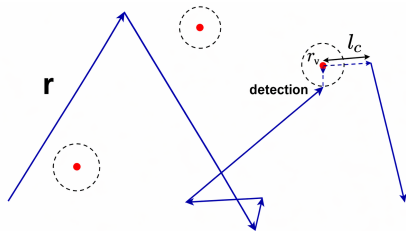


Figure: The Lévy walk search model and its parameters (Levernier, N).

Simulation setup

Model

- Targets: homogeneous PPP, density ρ
- Flights: $p(l) \propto l^{-\mu}$, $l \in [l_{\min}, l_{\max}]$
- Detection: corridor of width $2r_v$
- Mean free path: $\lambda = 1/(2\rho r_v)$
- Sparsity: $\lambda/r_v = 1/(2\rho r_v^2)$

Efficiency metric

$$\eta = \frac{\sum_r N_{\text{unique}}^{(r)}}{\sum_r D_{\text{total}}^{(r)}}$$

Pooled (global) estimator — robust to heavy-tailed per-run fluctuations.

Normalized: $\lambda\eta$ (dimensionless).

λ/r_v	ρ	Steps/run	Runs (destr.)	Runs (non-d.)
10–50	0.04–0.2	20k–30k	100	300
100–500	0.004–0.02	50k–80k	120	360
1000–2000	0.001–0.002	100k–150k	120	360

Phase diagram: $\mu^*(\lambda/r_v)$

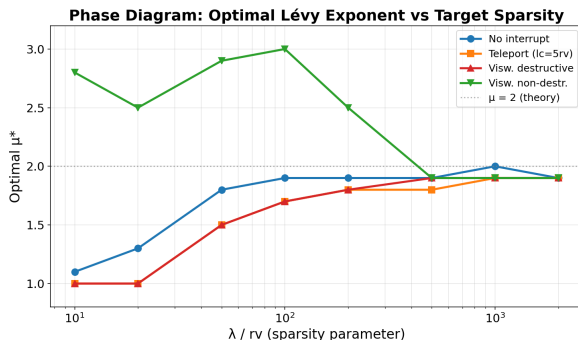


Figure: Optimal Lévy exponent μ^* vs. sparsity for all four strategies.

Key findings:

- $\lambda/r_v \geq 500$: all strategies $\rightarrow \mu^* \approx 2.0$ ✓
- $\lambda/r_v = 10\text{--}20$: three strategies shift toward $\mu^* = 1.0\text{--}1.4$ (ballistic)
- Non-destructive Viswanathan: $\mu^* \approx 2.2\text{--}2.6$ at high ρ (trapping artifact)

$\mu^* \approx 2$ confirmed in sparse limit.
At high density: ballistic is optimal.

Phase diagram: $\mu^*(\lambda/r_v)$ — with uncertainty

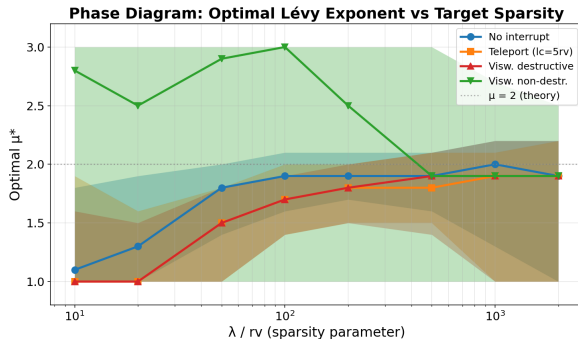


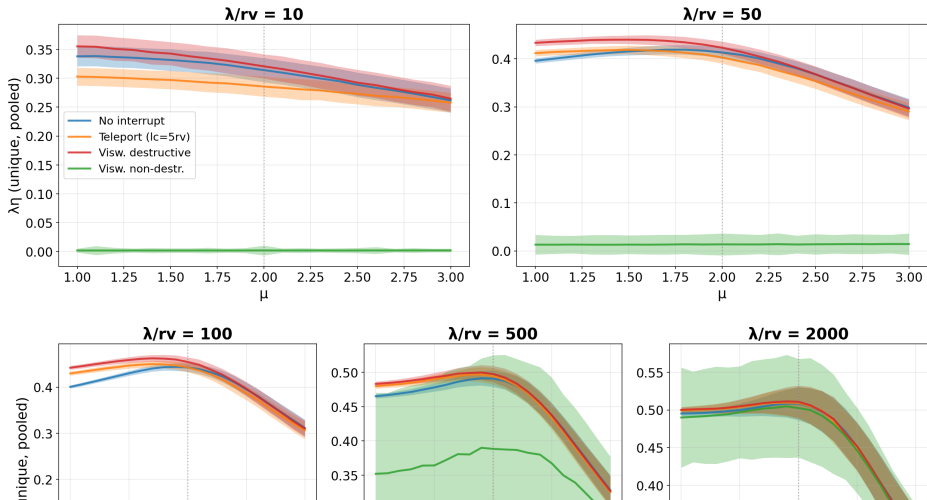
Figure: Same data with $\pm 1\sigma$ uncertainty bands.

Uncertainty bands:

- Shaded regions: all μ where $\eta(\mu) \geq \eta(\mu^*) - \sigma$.
- At $\lambda/r_v \geq 100$: narrow bands $\rightarrow$ well-determined μ^* .
- At $\lambda/r_v = 10$: wide bands $\rightarrow$ exponent matters less.

Efficiency curves $\lambda\eta(\mu)$

$\lambda\eta(\mu)$ Curves at Different Sparsity Levels



Mean first passage time (MFPT)

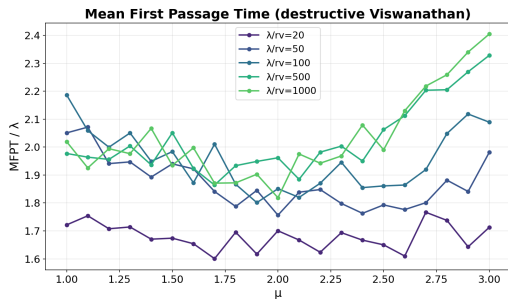


Figure: MFPT/ λ vs. μ (destructive Viswanathan).

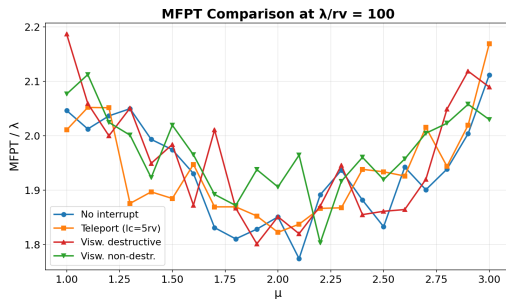


Figure: MFPT comparison at $\lambda/r_v = 100$.

- MFPT minimum near $\mu \approx 1.5$ at $\lambda/r_v = 100$ — lower than the throughput optimum ($\mu^* \approx 1.7$).
- Non-destructive Viswanathan: highest MFPT due to trapping.

Comparison with Previous Work

Our results at $\lambda/r_v \geq 500$ confirm the central prediction of Viswanathan et al. (1999), with $\mu^* = 1.8\text{--}2.0$ across all four strategies. The qualitative observation by Bartumeus et al. (2005) that the Lévy advantage weakens at high density is now quantified: the crossover from Lévy-optimal ($\mu^* \approx 2$) to ballistic-optimal ($\mu^* \rightarrow 1$) occurs at $\lambda/r_v \approx 50\text{--}200$, depending on the strategy.

The scaling critique of Levernier et al. (2020), that in $d \geq 2$ the exponent affects only prefactors, is consistent with our observation that $\lambda\eta$ varies by only a factor of ~ 2 across $\mu \in [1, 3]$ even in the sparse limit. The non-destructive Viswanathan results (with effective $l_c = 0$) show trapping pathology rather than the clean optimum predicted by Buldyrev et al. (2021), highlighting extreme sensitivity to the restart mechanism.

Summary and Conclusions

As a result of our work, we can draw the following conclusions. The optimal Lévy exponent converges to $\mu^* \approx 2$ in the sparse-target limit ($\lambda/r_v \geq 500$) for all four strategies, confirming the Lévy foraging hypothesis under the conditions where it was originally formulated. However, at high target density ($\lambda/r_v \leq 20$), the optimum shifts to $\mu^* \rightarrow 1$ (ballistic) for all strategies except non-destructive Viswanathan, where “ping-pong” trapping ($\sim 10^3$ revisits per unique target) creates a spurious shift of μ^* above 2.

The MFPT-optimal exponent ($\mu \approx 1.5$) is shifted toward ballistic values compared to the throughput optimum ($\mu^* \approx 1.7$), confirming that these two objectives select for different strategies. The standard truncation $l_{\min} = r_v$, $l_{\max} = \lambda$ is robust, with μ^* insensitive to moderate variations of these bounds.

Possible Future Directions

The present study suggests several natural extensions. A complete MFPT phase diagram across all sparsity levels and strategies would clarify whether the throughput–MFPT gap persists at extreme sparsity. Extending the simulation to three dimensions would test whether quantitative crossover densities change, since the detection geometry (cylinder vs. corridor) introduces new scaling. Replacing the Poisson field with clustered distributions would probe the robustness of the phase diagram against realistic spatial correlations.

Varying the restart distance l_c from 0 to $\gg r_v$ would map the full phase diagram in the $(l_c, \lambda/r_v)$ plane. Incorporating energy-aware metrics could shift the optimum at high density, and comparing with reinforcement-learning agents Muñoz-Gil et al. (2023) would quantify how close the power-law family is to the true optimum.

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Thank you!

Appendix: Convergence analysis

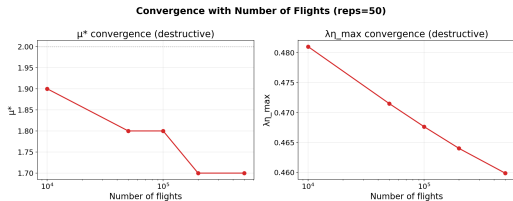


Figure: Convergence in number of flights (120 reps).

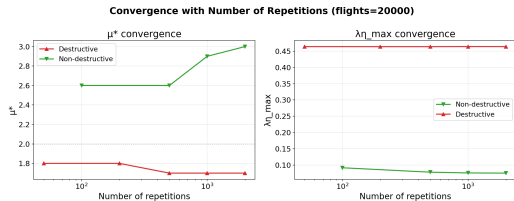


Figure: Convergence in number of runs (50k flights).

- Destructive: μ^* stable from $N_{\text{flights}} \geq 5000$, $R \geq 50$.
- Non-destructive: requires $R \geq 200$ due to trapping outliers.
- Production runs are well-converged ($< 1.5\%$ change).

Appendix: Analytical validation — $\eta = 1/(\langle l \rangle \cdot N)$

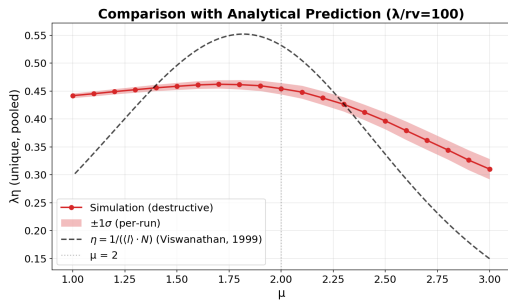


Figure: Simulation vs. Viswanathan (1999) at $\lambda/r_v = 100$.

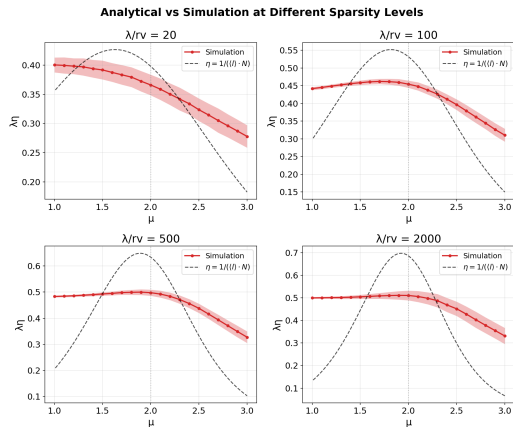


Figure: Analytical vs. simulation at four sparsity levels.

Appendix: μ^* values across all densities

λ/r_v	No interrupt	Teleport	Visw. destr.	Visw. non-d.
10	1.1	1.0	1.0	2.2
20	1.4	1.4	1.2	2.6
50	1.8	1.8	1.7	2.6
100	1.9	1.9	1.7	2.5
200	1.9	1.8	1.9	2.0
500	2.0	1.9	1.8	1.9
1000	2.0	2.0	1.9	2.0
2000	2.0	2.0	1.8	2.0

Table: Optimal μ^* at each λ/r_v . Resolution: $\Delta\mu = 0.1$.

Appendix: Trapping effect (non-destructive Viswanathan)

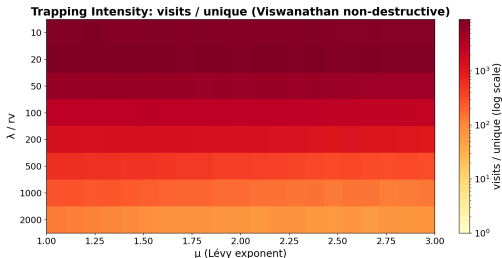


Figure: Trapping ratio V/U (visits per unique capture). Log scale.

Mechanism: forager restarts from target
→ short flight → re-detects nearby target
→ “ping-pong” cycle.

λ/r_v	V/U ($\mu=1$)	V/U ($\mu=2$)
10	1540	1716
50	43	53
100	11	14
500	1.6	2.0
2000	1.0	1.2

At $\lambda/r_v = 10$: each unique target revisited
~1700 times.

At $\lambda/r_v \geq 500$: trapping vanishes.

Appendix: Sensitivity to truncation parameters

Sensitivity to Power-Law Truncation (Destructive Viswanathan, $\lambda/r_v=100$)

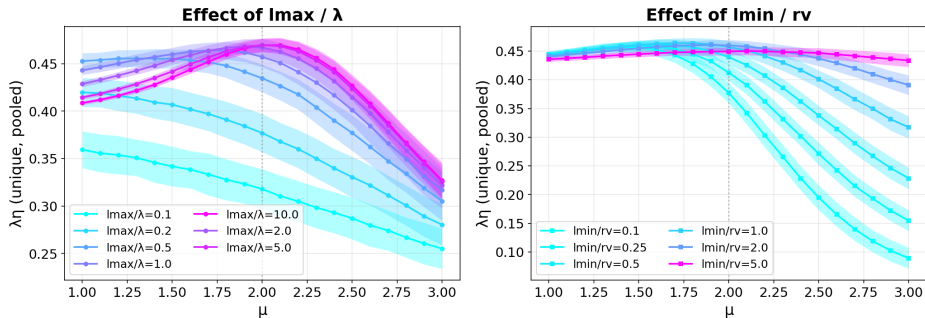


Figure: $\lambda\eta(\mu)$ for varying $l_{\max}/\lambda$ (left) and $l_{\min}/r_v$ (right). Shaded bands: $\pm 1\sigma$. The peak position μ^* is robust for $l_{\max}/\lambda \geq 1$ and $l_{\min}/r_v \leq 2$.